Conclusion: Eleven Formal and Computational Contributions

What This Paper Actually Did: Eleven Concrete Deliverables

This paper developed a unified category-theoretic framework for linguistic case systems, synthesizing five research traditions—typological, type-logical, distributional, enriched-categorical, and topos-theoretic—and embedding the result within an active inference model of cognition. Our eleven principal contributions are:

> ***Notation***: *Each entry is labelled* ***C_n_*** *(nth contribution,* ***C1–C11****) or* ***F_n_*** *(nth future direction,* ***F1–F7****).*

C1: Case Categories as a Formal Algebraic Framework

The review formalized case systems as categories with case roles as objects and grammatical relations as morphisms (**??**). This formalization captures the full typological range of alignment systems—nominative-accusative, ergative-absolutive, active-stative, tripartite, and fluid-S—within a single algebraic framework, with alignment functors providing structure-preserving mappings between systems. By tracing the lineage from Fillmore's

Seven Open Directions

The following seven directions (**F1–F7**) identify the most tractable and impactful extensions of this framework:

F1: Computational Experiments with DisCoCirc and lambeq

The DisCoCirc framework [@defelice2022discocirc] offers a natural platform for testing our case-theoretic predictions computationally. The release of **lambeq Gen II** [@lambeq2025genii] with full DisCoCirc support makes this direction immediately tractable: discourse-level case role tracking—including the dynamic role reversals of **??**—can now be compiled into parameterized quantum circuits (PQCs) and trained end-to-end. Krawchuk et al. [@krawchuk2025paramqnlp] demonstrate efficient generation of PQCs from large-scale texts (up to 6,410 words) with competitive performance on sentiment classification; [@letcher2024tight] and [@rad2024trainability] provide gradient bounds and reduced-domain initialization techniques that mitigate barren plateaus, making discourse-level circuit training practically feasible. Concrete experiments could include:

- Implementing case-marked DisCoCat models in **lambeq Gen**

Case Categories Are the Geometry of Meaning: A Unifying Coda

The commutative diagram is the central motif of this review—both as a mathematical tool and as a cognitive instrument. The analysis has demonstrated that the same diagrammatic language that makes category theory effective for formalizing case systems also makes it effective for *thinking about* case systems: the spatial structure of a commutative diagram encodes relational information in a format that supports rapid search, pattern recognition, and free-ride inference.

This convergence of mathematical utility and cognitive efficacy is not accidental. If the active inference framework is correct, then the brain operates by constructing and updating generative models of the world's relational structure. Category theory provides the structural algebra for these generative models; commutative diagrams supply their natural topology; and case categories instantiate the precise relational vocabulary that cognitive systems deploy to organize experience into coherent narratives of *who does*